Causality and Machine Learning:
Causal Models

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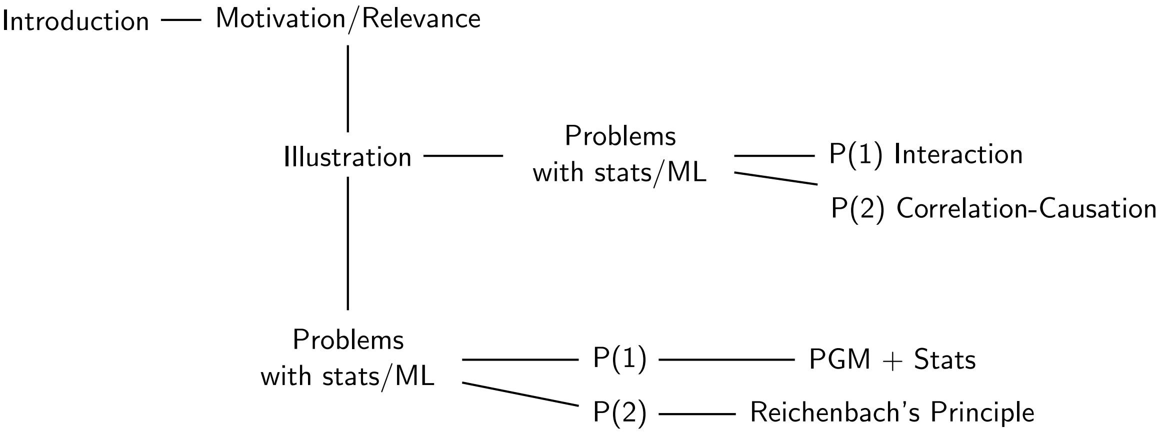
Objectives

- *Understand why causal reasoning is different from associational reasoning*
- *Distinguish different levels of causal reasoning*
- *Define models helping us perform causal reasoning*
- *Appreciate the challenges and advantages of performing causal reasoning*

1. Introduction

Roadmap



Introduction — Motivation/Relevance

Illustration — Problems with stats/ML

- P(1) Interaction
- P(2) Correlation-Causation

Problems with stats/ML

- P(1) — PGM + Stats
- P(2) — Reichenbach's Principle

1.1. Motivation

Talking about causality

We all talk about causality in our everyday life and in scientific research.

Smoking leads to tar deposits in the lungs, which in turn give rise to lung cancer.

We all intuitively understand the meaning of a similar sentence using words such as *leads*, *give rises*, *is at the root of*, *causes*...

Why do we talk about causality?

Causality is a very powerful concept.

Theoretically:

- It is the foundation of our understanding of the world.
- It is at the core of scientific endeavours.

Practically:

- It is effective for controlling the world.
- It allows us to differentiate association and direct relation.
- It provides a criterion to choose among models.
- It allows us to define parsimonious, robust and flexible models.

How can we talk about causality?

If causality is so useful, we want a *rigorous formalism* to reason about it.

- Can we use standard ML/statistical language?
- Do we need something new?

1.2. Illustration

Illustration: Ice Creams and Thefts

The city of Oslo has monitored the number of *ice-creams sold* (Ice) and the number of *thefts* (Thf) in town:

Ice	Thf
36	20
35	18
101	31
17	12
50	23
65	25
...	...

The administration has provided us with the data, asking to *"learn from the data"*.

Illustration: The Ideal Statistician

- ✓ We learn the *joint distribution* of the variables: $P(Ice, Thf)$
- ✓ We can *marginalize* and *condition*: $P(Thf), P(Thf|Ice)$

Ice	Thf
36	20
35	18
101	31
17	12
50	23
...	...

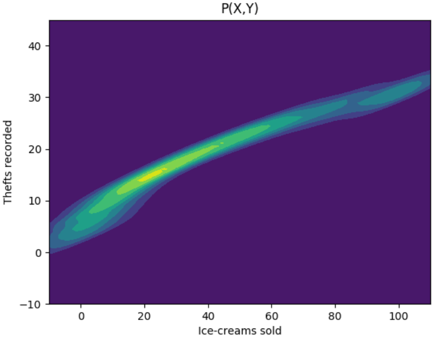
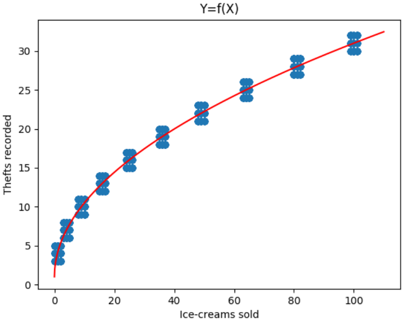


Illustration: The Ideal Machine Learner

- ✓ We can learn how the variables are *correlated*: $Ice \uparrow$, $Thf \uparrow$
- ✓ We can *predict* a variable from another: $Thf = f(Ice)$, $Ice = f(Thf)$

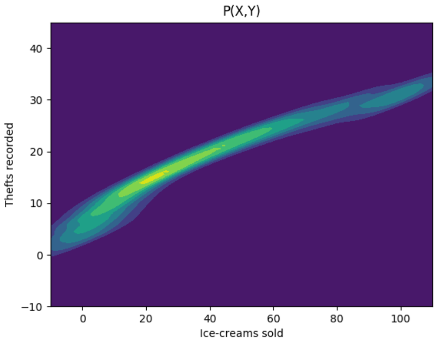
Ice	Thf
36	20
35	18
101	31
17	12
50	23
...	...

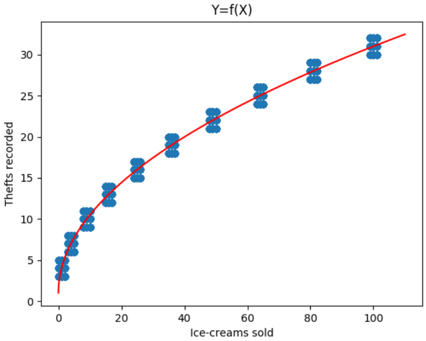


$$\boxed{Thf = 3 * \sqrt{Ice} + 1}$$

Illustration: The Control Problem

Now the administration wants to make use of these models:





A council member suggests: *Great work! Now that we know this monotonic relation we can act on the ice-cream parlours to reduce the rate of crime!*

Discussion

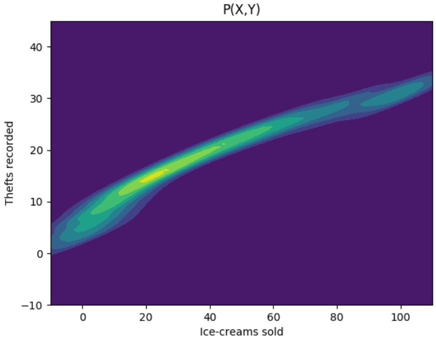
A council member suggests: *Great work! Now that we know this monotonic relation we can act on the ice-cream parlours to reduce the rate of crime!*

- What is your take on this suggestion?
- What is your understanding as a ML/stats advisor?
- What would you expect from the suggested policy?

Discuss these questions in groups.

Illustration: The Naive Statistician

Let's compute the conditional for $Ice = 0$.



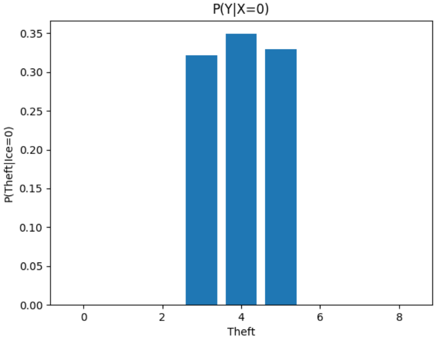
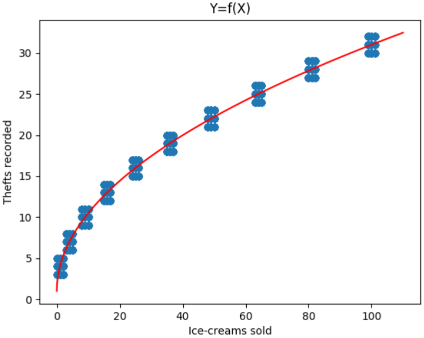


Illustration: The Naive Machine Learner

Let's use our model to compute $Ice = 0$.



$$\boxed{Thf = 3 * \sqrt{Ice} + 1}$$

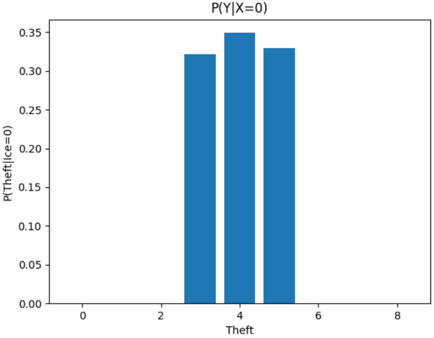
$$Thf = 3 * \sqrt{0} + 1$$

$$Thf = 1$$

Illustration: Clashing with Reality

Checking against reality might be disappointing:

The naive answers:



$$\boxed{Thf = 1}$$

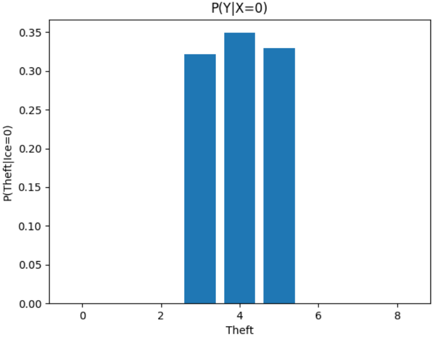
Collected data:

Ice	Thf
0	6
0	29
0	9
0	10
0	17
0	12
0	14
...	...

Illustration: Clashing with Reality

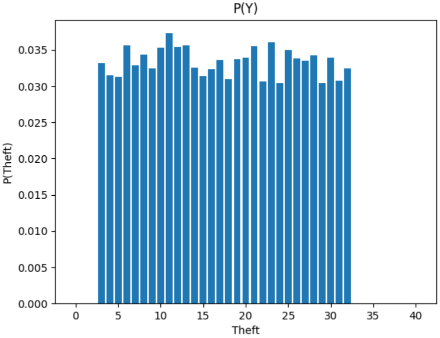
Checking against reality might be disappointing:

The naive answers:



$$\boxed{Thf = 1}$$

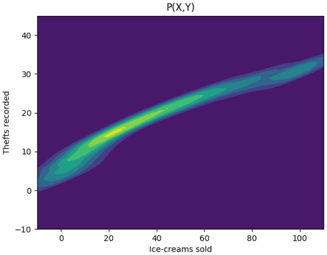
Collected data:



$$E[Thf] = 17.628$$

Illustration: What's the Problem in What We Did?

From the point of view of the *data model*:



- Changing *Ice* means changing the joint distribution.
- Samples are not from the same distribution anymore.

Illustration: What's the Problem in What We Did?

From the point of view of the *learned model*:

$$\boxed{Thf = 3 * \sqrt{Ice} + 1}$$

- The input-output relation is not causal.
- We learned to predict a correlation, not a causal mechanism.

Illustration: Is There a Solution?

We could dismiss this problem as a *distribution shift* and suggest either learning a new model or perform some degree of transfer learning.

Yet, we know that the dynamics of the system we observe are largely the same: we are still monitoring the same town.

There is some strong underlying (causal) **invariance** that our modelling has missed.

If we capture it exactly, we might be able to easily solve our discrepancy.

Recap: Key Problems

The illustration highlights two key problems in our way:

(P1) **Interaction problem:** The formalism we have been using can not express the idea of a policy artificially changing the value of a variable

(P2) **Association-Causality problem:** The formalism we have been using has not grounded way to discriminate between association and causality.

1.3. Towards a Causal Language

Defining causality

To proceed we need to define *causality*. But how to capture this idea properly?

Theoretically, very hard to define (Aristotle, Aquinas, Hume, Kirchoff, Russell, Rohrlich, Lewis... [2])

Many fields have dealt with this idea (philosophy, statistics, econometrics, epidemiology, social psychology, computer science)

Many approaches have been proposed in the recent years: randomization, potential outcomes (Neyman, Rubin), structural causal models (Pearl, Halpern, Dawid), single-world intervention graphs (Richardson, Rubin), counterfactual logic (Brigg).

Operational Approach

We will rely instead on our **operational** approach that allows us to solve the two key problems identified before:

(P1) **Interaction problem:** The formalism we have been using can not express the idea of variables having relations among them, and experimenters being able to change them.

(P2) **Correlation-Causality problem:** The formalism we have been using has not grounded way to discriminate between association and causality.

We want a framework that allows us to solve these two issues operationally - with minimal, *ontological* commitment.

(P1): Interaction Problem

Let us refine our definition of the **interaction problem** highlighting aspects of causality we miss:

1. It implies a *relationship* between things/variables.
2. We can probe causality by *interventions*.
3. It has a *counterfactual* aspect: *ceteribus paribus*, the presence or absence of a variable determines the outcome.

(P1): Causal Questions

In other words we want to be able to express the following questions:

1. **Associational questions**
   How are thefts and ice-cream related?
   Illustration showed that stats/ML are enough.
2. **Interventional questions**
   How do thefts changes if I prohibit ice-cream sales?
   Illustration showed that stats/ML here are struggling.
3. **Counterfactual questions**
   What would have happened to thefts if I had not prohibited ice-cream sales when instead I actually did?
   Illustration does not even consider how stats/ML would deal with such an individual-level question.

Discussion

Let us assume that causality describes **relations of cause-and-effect in the real-world** that allows us to reason and answer the questions we listed.

- What are the most basic features of these relations?
- What is formal/language features is stats/ML missing?

Discuss these questions in groups.

(P1): Key Features of Causal Interactions

Let us highlight *key features* of causal interactions understood as **relations of cause-and-effect in the real-world**.

- **Asymmetry**: causality implies directed/asymmetric relations between things;
- **Locality**: Causality defines local and sparse relation;
- **Autonomy**: Causal relations are autonomous and independent of each other;
- **Uncertainty**: Real-world causal relations are noisy/uncertain.

What *formalism*/*language* captures all these features?

(P1): Language of science

Now, is the standard mathematical language of science fit for causality?

$$\begin{aligned} e^{i\pi} &= -1 \\ F &= ma \\ V &= RI \\ y &= f(x) \end{aligned}$$

None of these are formally causal relations.

- × Symmetric
- × No notion of locality
- × No notion of autonomy
- × Deterministic

(P1): Language of statistics

How about the language of statistics?

$$P(A|B) = \frac{P(B|A)}{P(A)}$$

$$y = \beta_1 x + \beta_2 + \eta$$

$$y = f(x) + \eta$$

- × Symmetric
- × No notion of locality
- × No notion of autonomy
- ✓ Probabilistic

(P1): Limits of the statistical language

Indeed, from our illustration before, we have already seen that there are ideas in causality that are not well captured in standard scientific language of statistics/ML:

Causality	**Statistics**
Directionality	Non-directionality
Cause	Association
Causation	Correlation
Intervention	Observation
Action	Prediction

We experienced a **chasm between causality and statistics**.

(P1): Desiderata for a New Language

We want a **language** that allow us to express the ideas and questions we care about (*cause*, *intervention*, *counterfactual*).

We want a **theory** that bridges the gap with statistics.

causality formalism ⟶ statistical formalism

interventional domain ⟶ observational domain

(P1): Blueprint for a Causal Language

We design a language that from the start accounts for:

- ✓ *Asymmetry* ⇒ Directed arrows
- ✓ *Locality* ⇒ Graphs
- ✓ *Autonomy* ⇒ Mechanistic functions
- ✓ *Uncertainty* ⇒ Probabilities

We get all these requirements by imposing a *causal semantics* over the union of:

probabilistic graphical models with **statistics**.

(P2): Correlation-Causation Problem

We still have the fundamental problem of **correlation and causation**:

> How is it that things are related, but sometimes this relationship is a correlation and sometimes is causation?

We need to answer this question to really understand correlation and causation.

Discussion

We all know the ML mantra *"correlation is not causation"*.

- How do you distinguish the two?
- How do you formally express these two different relations?
- Are they related? How are they related?

Discuss these questions in groups.

(P2): Reichenbach's Principle

Reichenbach's Principle: two correlated variables X and Y can be causally related in only three ways[1]:

- X causes Y ($X \rightarrow Y$);
- Y causes X ($X \leftarrow Y$);
- X and Y have a common cause ($X \leftarrow Z \rightarrow Y$);
- A combination of the above.

Everytime we observe a correlation, there is causation at play - the problem is deciding what causal structure is underlying the data.

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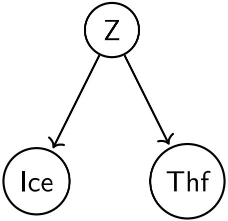
[1] Excluding colliders and coincidences.

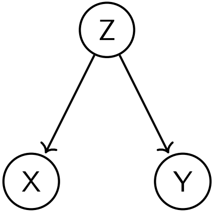
(P2): Back to illustration

Between ice-cream sold annd thefts, there was likely a *common cause* (Z), something such as season or temperature:



(P2): Common Causes Jargon

A *common cause* is often called a **confounder**, because it confounds the relationship between variables.



We say that:

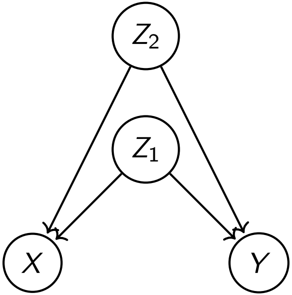
- “a confounder **explains** a relationship”
- “we **control** for a confounder”

to express the need to take a confounder into account while studying causal relatioships.

(P2): Confounders

Confounders are the *bugbear* of causal analysis.

- Is there a counfounder between the variables?
- Is the confounder I am observing completely explaining the causal relationship?
- Are there more than one confounder?
- Am I observing all the confounders?

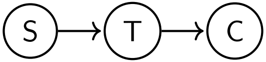


Recap: DAGs

(P1) *Graphical models* are a natural and intuitive way to express causal relatioships and capture their features;

(P2) *Graphical models* naturally allow us to discriminate correlation and causality;

Directed acyclic graphs are a natural and intuitive way to express causal relatioships and their directionality.

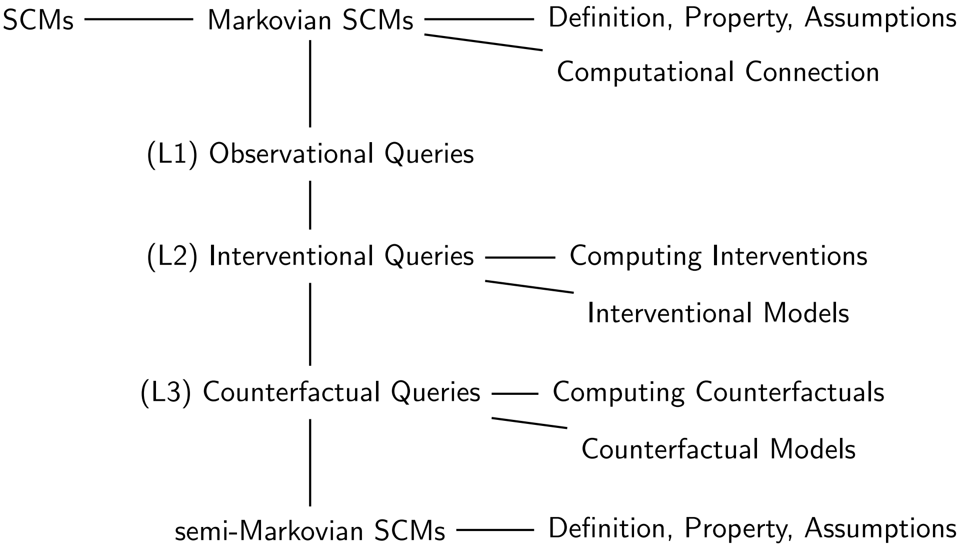


Directed acyclic graphs are a purely *mathematical structure* which:

- We have now endows with *causal meaning*.
- We will next enrich with *functions* and *distributions*.

2. Structural Causal Models

Roadmap



SCMs — Markovian SCMs

Markovian SCMs — Definition, Property, Assumptions

Markovian SCMs — Computational Connection

(L1) Observational Queries

(L2) Interventional Queries — Computing Interventions

(L2) Interventional Queries — Interventional Models

(L3) Counterfactual Queries — Computing Counterfactuals

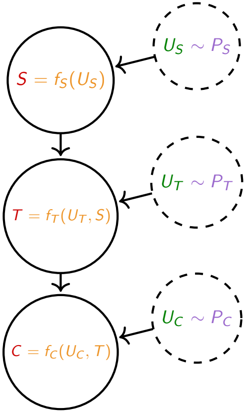
(L3) Counterfactual Queries — Counterfactual Models

semi-Markovian SCMs — Definition, Property, Assumptions

2.1. Markovian SCMs

SCMs: Definition

We express a **SCM** as $\mathcal{M} = \langle \mathcal{X}, \mathcal{U}, \mathcal{F}, \mathcal{P} \rangle$ [4, 6]:



- $\mathcal{X}$: set of *endogenous nodes* (S, T, C) representing variables of interest
- $\mathcal{U}$: Set of *exogenous nodes* (U_S, U_T, U_C) representing stochastic factors
- $\mathcal{F}$: Set of *structural functions* (f_S, f_T, f_C) describing the dynamics of each variable
- $\mathcal{P}$: Set of *distributions* (P_S, P_T, P_C) describing the random factors

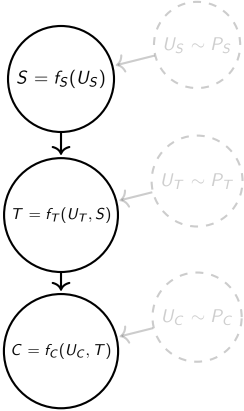
Such that the implied DAG is *acyclic*.

SCM: Features

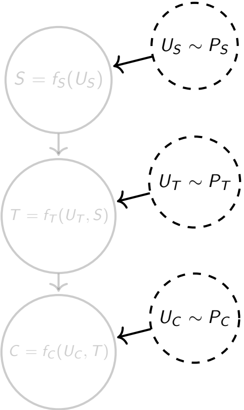
An SCM complies with the requirements for modelling causality:

- ✓ *Asymmetry*: defined by input and output of the acyclic set $\mathcal{F}$ of structural function;
- ✓ *Locality*: defined by nodes affecting only their children;
- ✓ *Autonomy*: defined by each structural function to be defined independently;
- ✓ *Uncertainty*: defined by the probability distributions in the set $\mathcal{P}$.

SCM: Determistic/Stochastic Dichotomy

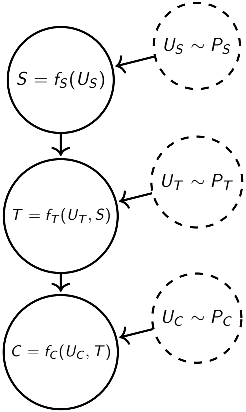


Endogenous nodes and structural functions are completely **deterministic**.



Exogenous nodes and probability distributions are completely **stochastic**.

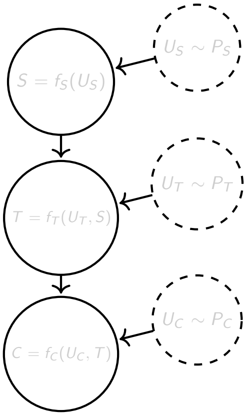
SCM: Re-composing the Determistic/Stochastic Dichotomy



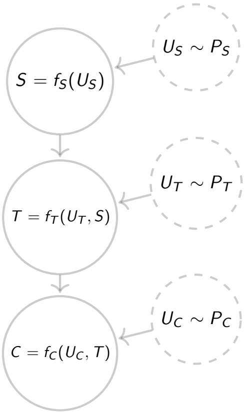
- *Empirically*, we do *ancestral sampling* by sampling exogenous variables and then computing endogenous variables.
- *Formally*, we *push-forward* the joint distribution over the exogenous variables over the endogenous variables.

At the end, we have a joint **distribution** $P_{\mathcal{M}}(S, T, C)$ explaining the stochastic behaviour of a deterministic system.

SCM: Graphical/Functional Dichotomy

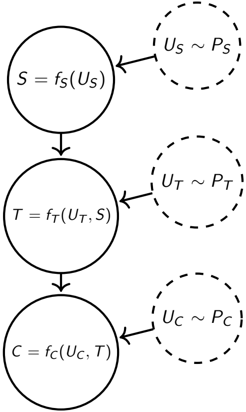


Graphical aspect determines the structure: what affects what.



Functional aspect determines the quantitative relation: how something affects something else.

SCM: Re-composing the Graphical/Functional Dichotomy



- **Markovianity** connects properties of the graph (d-separation) to properties of the data (independencies).
- **Faithfulness** connects properties of the data (independencies) to property of the graph (d-separation).

At the end, we can have an alignment between graphical and functional properties.

SCM: A recap of common assumptions

- *DAG level*
  - **Directionality**: edges have a direction.
  - **Acyclicity**: no loops in the graph.
- *BN level*
  - **Markovianity**: d-separation implies independence.
  - **Faithfulness**: independence implies d-separation.
- *CBN level*
  - **Causal arrows**: arrows represent causal relationships; missing arrow means no causal relationship.
  - **Causal relationship completeness**: all causes among the variables in the model are present.
  - **Causal sufficiency**: all common causes are modeled.
  - **Autonomy**: each variable is governed by an autonomous distribution.
- *SCM level*
  - **Non-parametric functions**: structural function are not constrained in their form.
  - **Autonomous functions**: each variable is governed by an autonomous function.
  - **Independent noise**: each variable has a single independent source of noise.

SCM: Important assumption equivalences

(CBN) *Causal sufficiency* $\equiv$ *No unobserved confounders* $\equiv$ *Independent noise* (SCM)

(CBN) *Autonomy* $\equiv$ *Autonomous function* (SCM)

(DAG) *Acyclicity* $+$ (SCM) *Independent noise* $\equiv$ *Markovianity* (BN)

SCM: Markovian SCMs

An important class of SCMs are **Markovian SCMs**, that is, SCMs that are:

- ✓ Acyclic
- ✓ Independent noise

These are the “easy models” of causality:

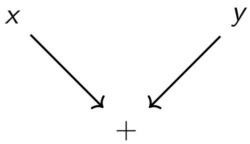
- ✓ There are no hidden confounders;
- ✓ Causal reasoning is possible.

SCM: Computational Language

We said that the language of stats/ML was insufficient for causality, so we came up with a new formalism based on graphs and statistics.

Still there is a formalism that already captures those properties we care about: **computation**.

$$
\begin{aligned}
x &\leftarrow 2 \\
y &\leftarrow \texttt{fun}(z)
\end{aligned}
$$



Assignments are *asymmetric* and programs define *computational graphs*.

SCM: Computation-Causality Connection

Programs satisfy the requirements we made of causality:

- ✓ *Asymmetry*: defined by input and output of programmatic functions;
- ✓ *Locality*: defined by relations among variables as specified in the computational graph;
- ✓ *Autonomy*: defined by each programmatic function to be defined independently;
- ✓ *Uncertainty*: defined by random number generation,

Causal system $\equiv$ **Data generating system** $\equiv$ **Program**

We will exploit this connection [3] to play programmaticaly with causality!

2.2. (L1) SCM and Observational Questions

(L1): Observational Queries

How do we use a SCM to answer an **observational query**, such as *what is the probability of theft observing that no ice cream is sold?*

1. Let's express the query formally:

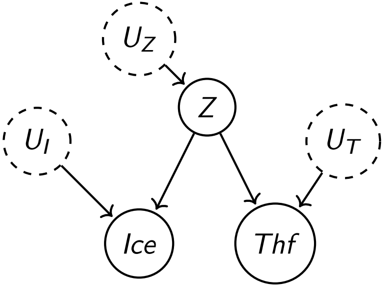
$$P(Thf|Ice = 0)$$

2. Let's recall the conclusion of the re-composition of the deterministic/stochastic dichotomy:

$$\text{there exist the joint } P(Thf, Ice, Z)$$

3. From the joint I can compute any marginal/conditional.

(L1): Illustration



What is the probability of Thf observing Ice=0?

$$P_{\mathcal{M}}(Thf|Ice=0)=$$
$$=\frac{P_{\mathcal{M}}(Thf,Ice=0)}{P_{\mathcal{M}}(Ice=0)}=$$
$$=\frac{\sum_Z P_{\mathcal{M}}(Thf,Ice=0,Z)}{\sum_{Thf,Z} P_{\mathcal{M}}(Thf,Ice=0,Z)}$$

Exercise

Implement in Python from scratch the following causal model:

> *We are studying a lung-cancer model focused on five variables. We have a binary* ENVIRONMENT *variable determined uniquely by the noise* $\texttt{Bern}(0.3)$*. Environment is affecting a binary* SMOKING *variable whose value is one if the sum of environment and a* $\texttt{Bern}(0.5)$ *noise is greater than one. Smoking feeds into another binary variable,* TAR DEPOSITS*, which achieves one if the difference between smoking and* $\texttt{Bern}(0.1)$ *is greater or equal to one. A key variable is* CANCER*, whose binary value is one if the sum of half tar, environment and half* $\texttt{N}(0,1)$ *is greater than 1.5. Finally we track a* COUGH *variable, equal to half the sum of smoking, cancer and* $\texttt{Bern}(0.4)$*.*

- Collect a sample by running the model
- Collect a dataset from the model
- Estimate/plot the following quantities: $P(Tar, Cancer)$; $P(Cancer|Smoking = 1)$

2.3. (L2) SCM and Interventional Questions

(L2): Interventional Queries

How do we use an SCM to answer an **interventional query**, such as *what is the effect on thefts of forcing ice cream sales to zero?*

1. Let's express the query formally:

????

After all our reasoning, it seems wise not to assume that this quantity has the same expression as $P(Thf|Ice=0)$.

Then what?

(L2): Defining an Intervention Operator

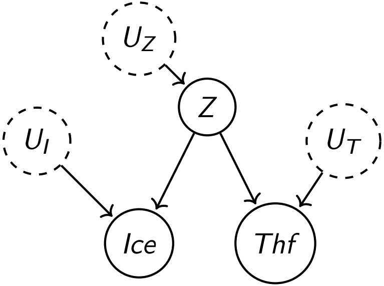
We define a new **intervention operator** $do(X = x)$ expressing that an experimenter intervenes on a system fixing the value of the variable X to x.

What does this mean/imply:

- An experimenter can act on a specific variable/function;
- An experimenter can affect *only* their target variable/function (granted by *autonomy* assumption of SCMs);
- An experimenter can change a target variable/function faultlessly (*perfect intervention* assumption);
- A target variable X is not defined by its mechanism $f_x(\mathtt{Pa}(X), U_X)$ but by the experimenter;
- A target variable X loses its causes $\mathtt{Pa}(X), U_X$ as the cause is now the experimenter;

(L2): Operationalizing an Intervention Operator

How does the intervention operator practically interact with the SCM?

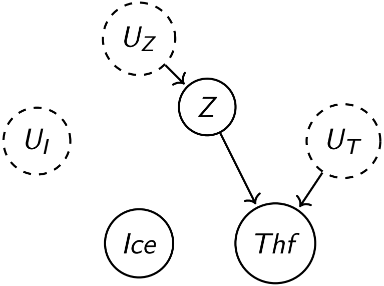


$do(Ice = 0)$

1.
2.

(L2): Operationalizing an Intervention Operator

How does the intervention operator practically interact with the SCM?

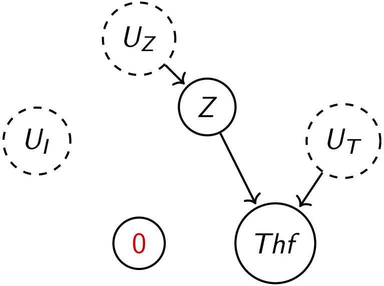


$do(Ice = 0)$

1. Remove incoming edges in the intervened node
   (i.e.: the intervened node loses its natural causes)
2.

(L2): Operationalizing an Intervention Operator

How does the intervention operator practically interact with the SCM?



$do(Ice = 0)$

1. Remove incoming edges in the intervened node
   (i.e.: the intervened node loses its natural causes)
2. Set the value of the intervened node
   (i.e.: replace the natural causal mechanism with the choice of the experimenter)

Exercise

Reconsider the causal model from the previous exercise:

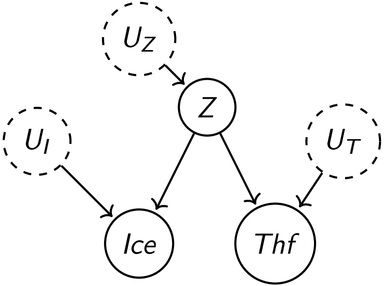
> *We are still studying the same lung-cancer model. We want to study the model under the intervention where patients are forced to smoke.*

- What is the implementation of an intervention on a program?
- Perform the intervention, and estimate/plot the following quantities: $P(Tar, Cancer)$; $P(Cancer|Smoking = 1)$.

(L2): Interventional Model

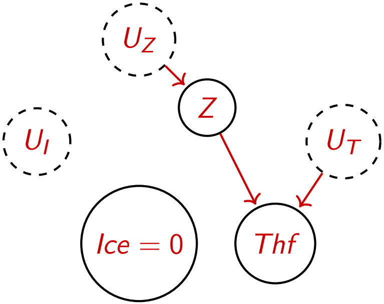
An *intervention* $\iota = do(Ice = 0)$ defines a new **intervened model** $\mathcal{M}_\iota$ with new distributions.

$\mathcal{M}$



$P_\mathcal{M}(Thf, Ice, Z)$

$\mathcal{M}_\iota$

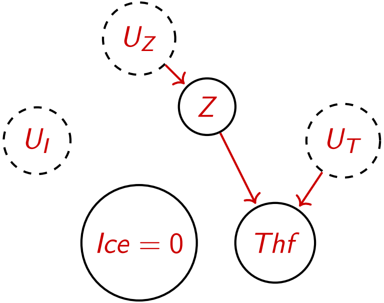


$P_{\mathcal{M}_\iota}(\text{Thf,Ice,Z})$

(L2): Interventional Models

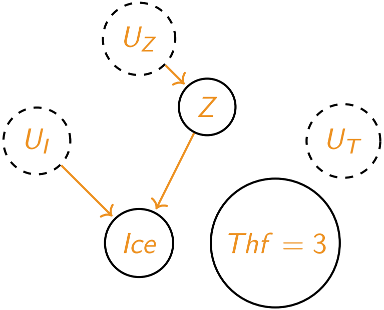
From an SCM, we can actually generate *multiple post-interventional models*, one for each intervention:

$\mathcal{M}_\iota$



$P_{\mathcal{M}_\iota}$(Thf,Ice,Z)

$\mathcal{M}_{\iota'}$



$P_{\mathcal{M}_{\iota'}}$(Thf,Ice,Z)

(L2): Interventional Models

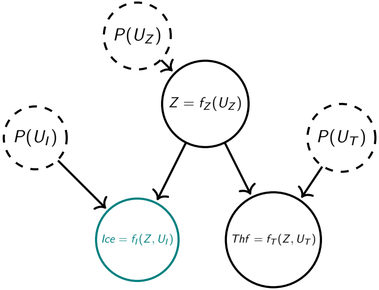
Interventions mean that an SCM is *not a static model*:

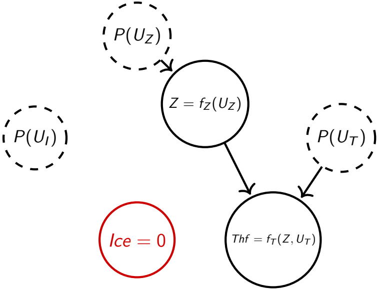
- An SCM is a **generator** of multiple graphical models explaining the dynamics of a system under different interventions;
- An SCM encode a **data-generating mechanism** which is **robust to external interventions**.

(L2) Conditioning is not Intervening

At this point, we should be convinced that:

Conditioning $\neq$ Intervention





$$P_{\mathcal{M}}\left(Thf \middle| Ice = 0\right)$$

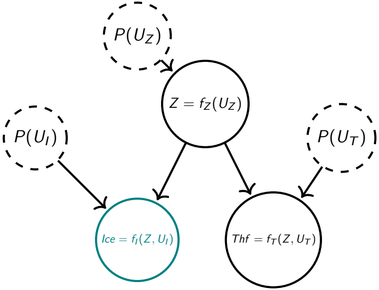
$$P_{\mathcal{M}}\left(Thf \middle| do\left(Ice = 0\right)\right)$$

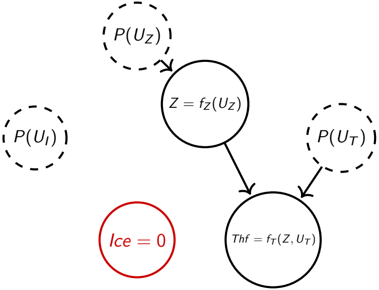
Knowledge of $Ice = 0$ allows inference on distribution of Z and then Thf.

Knowledge of $do\left(Ice = 0\right)$ does not affect the distribution of Z.

(L2) But Intervening is Conditioning in a Different Model

Although, we might still say that an *intervention* can be reduced to *conditioning* in the **induced post-interventional model**.





$$P_{\mathcal{M}}\left(Thf | Ice = 0\right)$$

$$P_{\mathcal{M}}\left(Thf | do\left(Ice = 0\right)\right)$$

$$P_{\mathcal{M}_{do}}\left(Thf | Ice = 0\right)$$

2.4. (L3) SCM and Counterfactual Quantities

(L3): Counterfactual Queries

How do we use an SCM to answer a **counterfactual query**, such as *what would have thefts been, had the sale of ice creams been zeroed when instead it was observed to be 42?*

1. Let's express the query formally:

$$????$$

Again, we face a shortage in language.

(L3): Defining Counterfactuals

A **counterfactual** is a possible outcome which we did not observe, which could have been under the same circumstances if something different had taken place.

What does this mean/imply:

- There is an *actual world* that we observe;
- We assume that things would have gone differently if we had taken an *alternative action*
- The alternative action happens in the same world, leading to a *counterfactual outcome*

(L3): Existence of Counterfactuals

In every situation, we will always observe an *actual outcome* - i.e., we never observe *counterfactual outcomes*.

Then, does it make any sense to worry about **countefactuals**?

Counterfactuals are still important because:

- they are essential in our reasoning and understanding of the world;
- they are the basis of important concepts like *regret* or *acocuntability*;
- they are key to some very definition of *causality*.

Discussion

Assume that: *On a sunny day, we have observed a person crossing the street as the traffic light turned yellow.*

Suppose now we want to evaluate: *Would have that person crossed if the light had turned red?*

How shall we evaluate this query wrt everything else that has not been specified? What day shall we consider?

- The same sunny day we observed
- Another randomly selected day, possibly a rainy day
- The average day

Discuss this question in groups.

(L3): Closest World

It is assumed that counterfactuals must be evaluated in the **closest world** possible:

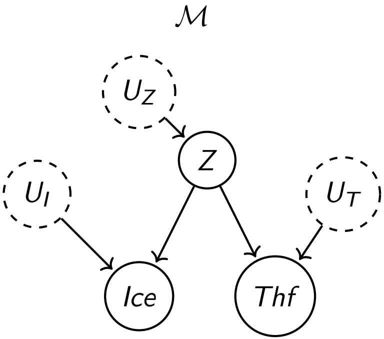
- The *minimally modified* world by our change
- Which is still *consistent*.

Hence, counterfactuals are by definition **individual quantities** not **population-level statistics**.

How do we find this *closest world possible*?

(L3): Back to Exogenous Variables

Let's go back to exogenous variables:



What is an exogenous variable:

- *To an engineer*: a source of noise;
- *To a doctor*: the state of a patient/unit;
- *To a philosopher*: the state of the world beyond our model;
- *To a computer scientist*: the context depending on random samples.

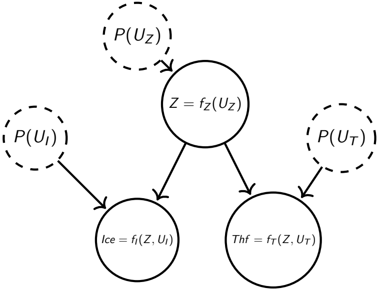
The key point is that a *setting of exogenous variables* can be seen as "the state of the world" that determines the *actual outcome* we observed.

(L3) Computing Counterfactuals

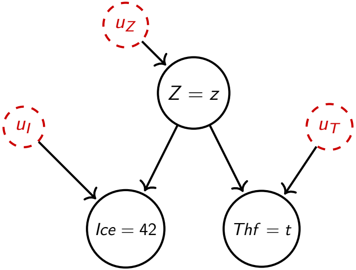
What would have Thf been, had Ice been set to 0 when instead it was observed to be 42?

$$P_{\mathcal{M}}(Thf_{Ice=42}|do(Ice=0))$$

1. *Abduction:* we start from the factual world, and infer the value/distribution of exogenous variables from observations.



(Original model)



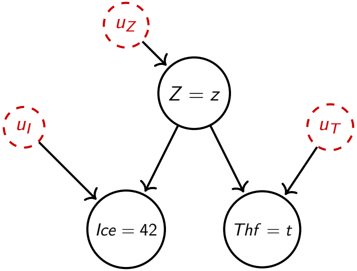
(Abduction)

(L3): Computing Counterfactuals

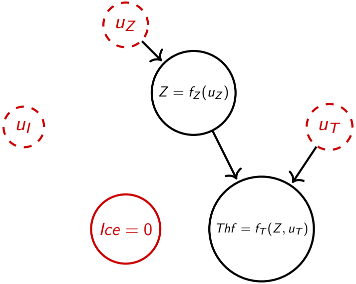
What would have Thf been, had Ice been set to 0 when instead it was observed to be 42?

$$P_{\mathcal{M}}(Thf_{Ice=42} | do(Ice = 0))$$

2. *Action:* intervene as requested in the counterfactual.



(Abducted model)



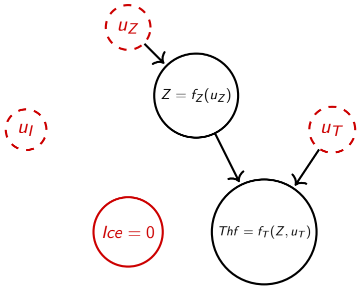
(Action)

(L3): Computing Counterfactuals

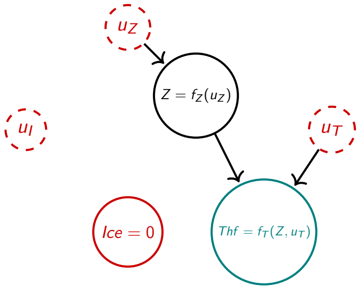
What would have Thf been, had Ice been set to 0 when instead it was observed to be 42?

$$P_{\mathcal{M}}(Thf_{Ice=42}|do(Ice=0))$$

3. *Prediction:* compute the variable of interest in the counterfactual model.



(Abducted-Acted model)



(Prediction)

Exercise

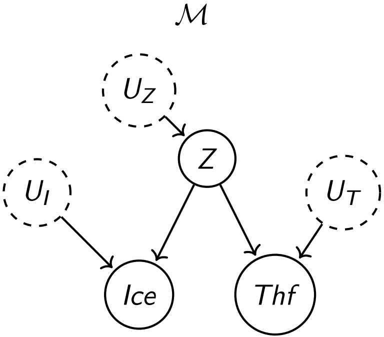
Reconcider the causal model from the previous exercise:

> *We are still studying the same lung-cancer model. After intervening to force each patient to smoke, we want to evaluate what would have been the individual outcomes if instead we had forced them not to smoke.*

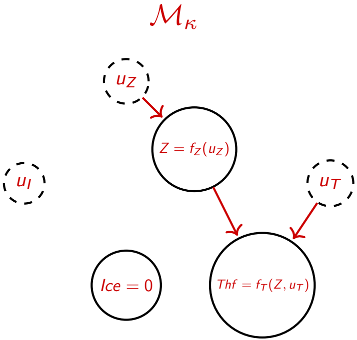
- What is the implementation of a counterfactual in a program?
- Implement the counterfactual, and estimate/plot the following quantities: $P_{do(S=1)}(Tar, Cancer|do(S=0))$; $P_{do(S=1)}(Cancer|do(S=0))$.

(L3): Counterfactual Model

As interventions, also a counterfactual $P_{\mathcal{M}}(Thf_{Ice=42} | do(Ice = 0))$ induces a new **counterfactual** model.



$P_{\mathcal{M}}(Thf, Ice, Z)$

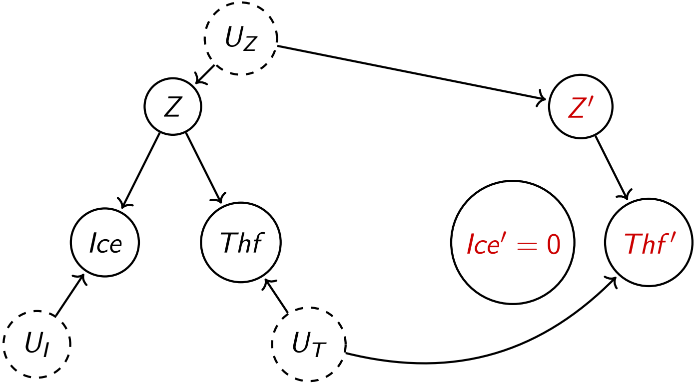


$P_{\mathcal{M}_\kappa}(Thf, Ice, Z)$

(being a singleton if abduction is perfect)

(L3): Twin Worlds

A more intuitive representations of a counterfactual is via a **twin model**:

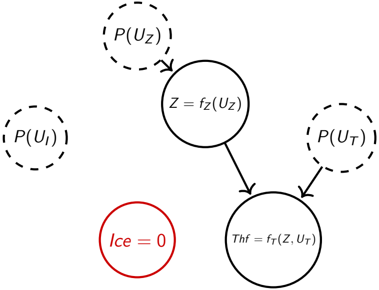


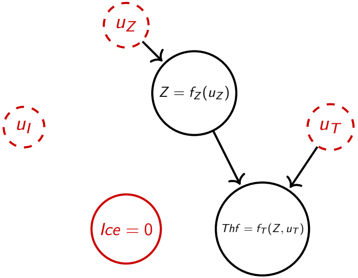
Connecting two models via *shared exogenous variables*.

(L3): Intervening is not a Counterfactual

At this point, we should be convinced that:

Intervention $\neq$ Counterfactuals





$$P_{\mathcal{M}}\left(Thf \,|\, do\left(Ice = 0\right)\right)$$

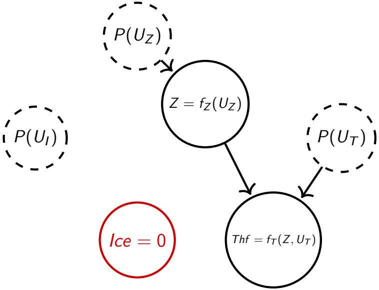
The value of *Thf* is independent of *Ice*, determined by a random samples of Z.

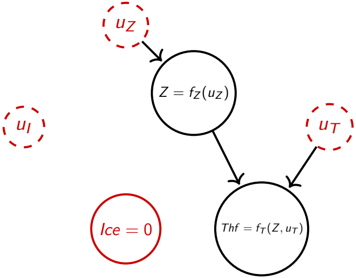
$$P_{\mathcal{M}}(Thf_{Ice=42} \,|\, do(Ice = 0))$$

The value of *Thf* is always independent of *Ice*, but the sample from Z is conditioned to the value that produced $Ice = 42$.

(L3) But Counterfactual is Intervening in a Different Model

Although, we might still say that a *counterfactual* can be reduced to *intervening* in the **induced counterfactual model**.





$$P_{\mathcal{M}}\left(Thf \middle| do\left(Ice = 0\right)\right)$$
$$P_{\mathcal{M}_{do}}\left(Thf \middle| Ice = 0\right)$$

$$P_{\mathcal{M}}(Thf_{Ice=42} | do(Ice = 0))$$
$$P_{\mathcal{M}_{Ice=42}}(Thf | do(Ice = 0))$$

Recap: Pearl's Causal Hierarchy

We can organize all the questions we have seen in the **Pearl's Causality Hierarchy** [1]:

Causality	**3. Counterfactuals**	What would have Y been, had X been x' when instead it was x? $P\left(Y_{do(X=x')} \middle| Y=y, X=x\right)$ Structural causal models
	2. Causal Effects	What is the effect of X on Y? $P\left(Y \middle| do\left(X=x\right)\right)$ Causal Bayesian networks
Stat/ML	**1. Associative Relationships**	How does Y relate to X? $P\left(Y \middle| X\right)$ Bayesian networks

Recap: Pearl's Causal Hierarchy

Causality	**3. Counterfactuals**	What would have Y been, had X been x' when instead it was x? $P\left(Y_{do(X=x')} \middle\vert Y=y, X=x\right)$ Structural causal models
	2. Causal Effects	What is the effect of X on Y? $P\left(Y \vert do\left(X=x\right)\right)$ Causal Bayesian networks
Stat/ML	**1. Associative Relationships**	How does Y relate to X? $P\left(Y \vert X\right)$ Bayesian networks

This is a **strict hierarchy**:

- Every level contains more information than the preceding level;
- A query at level $L(n)$ can not necessarily be reduced to a query at level $L(n-1)$;
- The hierarchy does not collapse.

2.5. Semi-Markovian SCMs

SM-SCM: Limits of Markovian SCMs

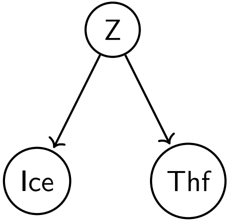
Recall that **Markovian SCMs** are the “easy models” of causality where there are *no hidden confounders*.

Unfortunately, in reality, it is very common to have *unobserved confounders.*

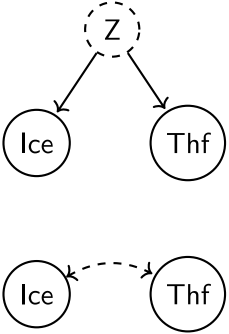
SM-SCM: Semi-Markovian SCMs

An more realistic class of SCMs are **semi-Markovian SCMs**, that is, SCMs that are:

- ✓ Acyclic
- ✓ No independent noise



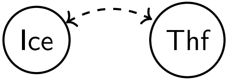
Markovian SCM



semi-Markovian SCM

SM-SCM: Semi-Markovian SCMs

A **semi-Markovian SCMs**:

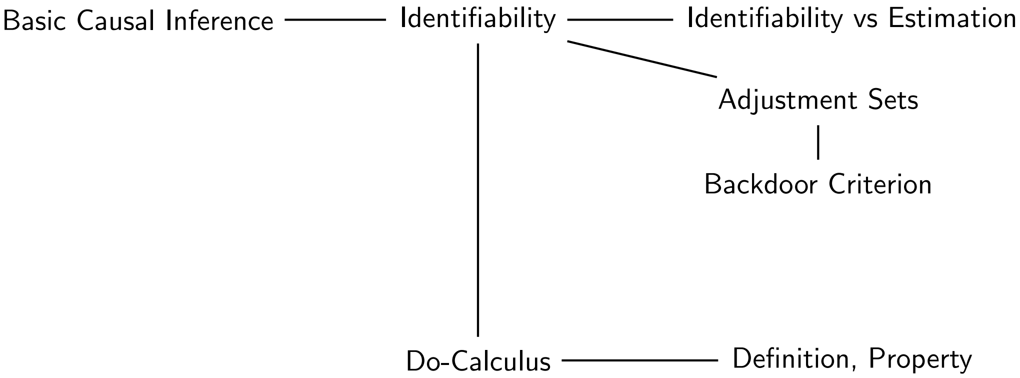


is an SCM that:

- It would reduce to a Markovian SCM *if we were to observe the hidden counfounders*;
- *without observing the hidden confounders* evaluating relationships of cause and effects is more challenging.

3. Basic Causal Inference

Roadmap



Basic Causal Inference

Identifiability

Identifiability vs Estimation

Adjustment Sets

Backdoor Criterion

Do-Calculus

Definition, Property

3.1. Identifiability

CI: Basic Causal Inference

So far:

- We have seen how to define causal models (*SCMs*);
- We have defined and computed observational, interventional and counterfactual quantities *given an SCM*.

If we had an SCM $\mathcal{M}$, we could easily compute **any** causal quantity.

- Just *solve* or *simulate* the SCM.

CI: Basic Causal Inference

In reality, normally we do not have a SCM. Instead we have:

- *A causal graph* $\mathcal{G}$
    - Just causal relationship between variables elicited from domain experts with no definition of structural functions or probability distributions.
- *Observational data* $\mathcal{D}^o$
    - Easier and cheaper to collect than interventional data.

Given $\langle \mathcal{G}, \mathcal{D}^o \rangle$, what causal queries can we answer?

CI: Identifiability and Estimation Problem

Let $\mathcal{Q}$ be a causal query. Given $\langle \mathcal{G}, \mathcal{D}^o \rangle$, can we answer $\mathcal{Q}$?

- **Identifiability problem**: has $\mathcal{Q}$ a unique answer computable from our data $\langle \mathcal{G}, \mathcal{D}^o \rangle$?
- **Estimation problem** (or *identification* problem): how do we compute the answer to $\mathcal{Q}$ from our data $\langle \mathcal{G}, \mathcal{D}^o \rangle$?

For the rest of today, we will discuss *identifiability*; *estimation* will be the topic of tomorrow.

CI: Identifiability Problem

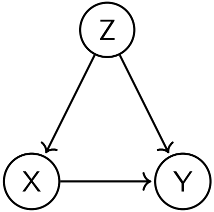
Given $\langle \mathcal{G}, \mathcal{D}^o \rangle$, what causal queries $\mathcal{Q}$ are identifiable?

- $\mathcal{Q} \in L1$: *observational queries* can be trivially answered since we have $\mathcal{D}^o$.
- $\mathcal{Q} \in L2/L3$: *interventional* and *counterfactual* queries are not trivially answered since PCH does not collapse.

CI: Identifiability of Interventional Queries

We now focus on **interventional queries** which convey essential causal statements.

Given $\mathcal{G}$



and $\mathcal{D}^o$

Can I identify a quantity such as

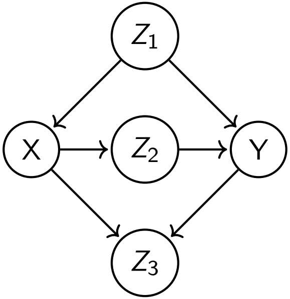
$$P(Y|do(X))$$

that is, the effect of X on Y?

CI: Identifiability of Interventional Queries

Notice a few challenges in dealing with *interventional queries*.

Given $\mathcal{G}$



and $\mathcal{D}^o$

- × Causal effects might be *confounded by spurious correlations* (Z)
- × Technically, we have no way of evaluating a quantity such as $P(Y|do(X))$ directly from $\mathcal{D}^o$.

Identifiability will deal with the question of whether we can transform a quantity such as $P(Y|do(X))$ into a *do-less* quantity (L1) that correctly estimates a causal effect from $\mathcal{D}^o$.

CI: Adjustment Set

A first intuition to evaluate a quantity like $P(Y|do(X))$ is to find an **adjustment set**[2] $\mathbf{Z}$ such that:

$$P(Y|do(X)) = \sum_{\mathbf{Z}} P(Y|X, \mathbf{Z})P(\mathbf{Z})$$

Such a set:

- ✓ Allows to *control for the confounder* (by stratification);
- ✓ Transforms an interventional query in a purely observational query estimable from $\mathcal{D}^o$.

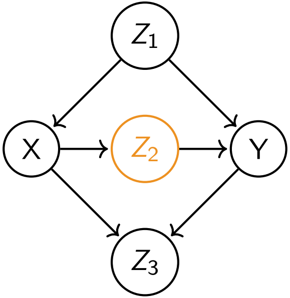
So, how do we determine the correct adjustment set?

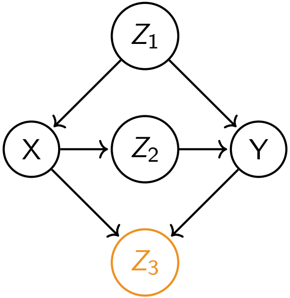
---

[2] a.k.a. *admissible set*, *deconfounding set*, *sufficient set*, *appropriate set*

CI: Adjustment Set

Given $P(Y|do(X))$ conditioning on everything does not seem a good idea:





If I were to condition on Z_2, I would **block a direct causal path** between X and Y.

If I were to condition on Z_3, I would **introduce selection bias** between X and Y.

We need a proper rigorous criterion.

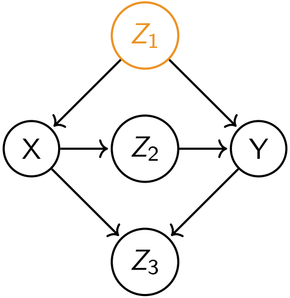
CI: Backdoor Criterion

The **backdoor criterion** states that a valid adjustment set $\mathbf{Z}$ is one that:

1. $\mathbf{Z}$ blocks every path between X and Y that has an arrow ingoing X;
2. $\mathbf{Z}$ contains no descendants of X;

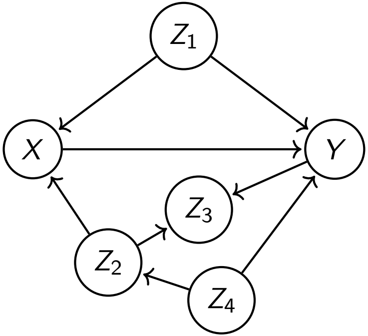
Intuitively, the criterion enforces:

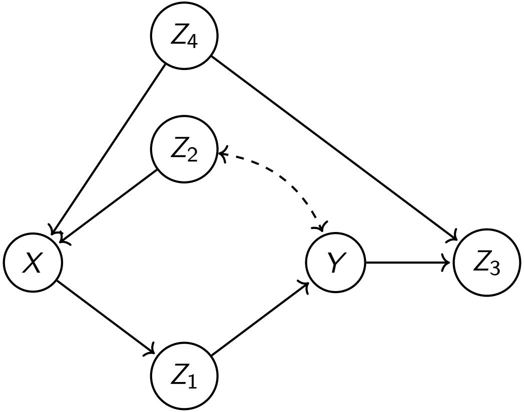
- No spurious path between X and Y is left open;
- No new spurious path is created;
- Directed paths between X and Y are left open.



Exercise

Apply the backdoor criterion to the following $\mathcal{G}$ wrt the effect of X on Y:

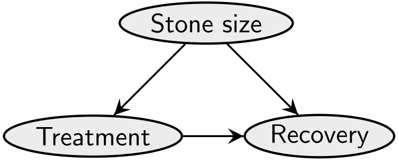




Exercise: Treatment of Kidney Stones

	Treatment A	Treatment B
Small stone	93% (81/87)	87% (234/270)
Large stone	73% (192/263)	69% (55/80)
Both	78% (273/350)	83% (289/350)

Table: Success rate of two types of treatment of large and small kidney stones.



3.2. Do-Calculus

CI: Beyond Backdoor

The backdoor criterion is *sound* (if we find an adjustement set we can use to identify an interventional query), but not *complete* (if we do not find an adjustment set, it might still be possible to identify an interventional query).

More, specialized criteria might be defined (e.g.: *frontdoor criterion*, *instrumental variables*...)

However, we would like a **generic method** to evaluate identifiability from $\langle \mathcal{G}, \mathcal{D}^o \rangle$...

CI: Do-Calculus

Amazingly, it turns out that there is a set of rules called **do-calculus**, that is *sound* and *complete* [8].

That is, given:

- An interventional query $\mathcal{Q}$
- A causal graph $\mathcal{G}$ (and observational data $\mathcal{D}^o$)

Repeated application of do-calculus rules (respecting $\mathcal{G}$) will reduce the interventional query $\mathcal{Q}$ to an observational query, if possible.

CI: Do-Calculus

There are three rules of **do-calculus** [5]:

1. *Rule for removing/inserting observations*

$$P(\mathbf{Y}|do(\mathbf{X}),\mathbf{Z},\mathbf{W}) = P(\mathbf{Y}|do(\mathbf{X}),\mathbf{W}) \quad \text{if } \mathbf{Y} \perp_d \mathbf{Z}|\mathbf{X},\mathbf{W} \text{ in } \mathcal{G}_{\bar{\mathbf{X}}}$$

2. *Rule for exchanging interventions and observations*

$$P(\mathbf{Y}|do(\mathbf{X}),do(\mathbf{Z}),\mathbf{W}) = P(\mathbf{Y}|do(\mathbf{X}),\mathbf{Z},\mathbf{W}) \quad \text{if } \mathbf{Y} \perp_d \mathbf{Z}|\mathbf{X},\mathbf{W} \text{ in } \mathcal{G}_{\bar{\mathbf{X}},\underline{\mathbf{Z}}}$$

3. *Rule for removing/inserting interventions*

$$P(\mathbf{Y}|do(\mathbf{X}),do(\mathbf{Z}),\mathbf{W}) = P(\mathbf{Y}|do(\mathbf{X}),\mathbf{W}) \quad \text{if } \mathbf{Y} \perp_d \mathbf{Z}|\mathbf{X},\mathbf{W} \text{ in } \mathcal{G}_{\bar{\mathbf{X}},\underline{\mathbf{Z}(\mathbf{W})}}$$

    - ($\mathbf{Z}(\mathbf{W})$ being subset of nodes in $\mathbf{Z}$ not ancestors of any node in $\mathbf{W}$).

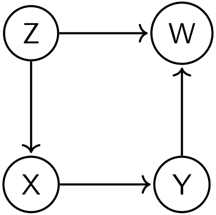
CI: Identifiability via Do-Calculus

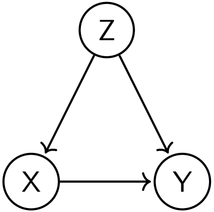
Important properties of **do-calculus**:

- It is **complete**: an interventional query $\mathcal{Q}$ is identifiable from $\langle\mathcal{G},\mathcal{D}^o\rangle$ iff it can be reduced to an observational query via do-calculus.
- Any interventional query $\mathcal{Q}$ in a Markovian SCM is identifiable.
- Interventional queries $\mathcal{Q}$ on a semi-Markovian SCM are not necessarily identifiable.
- The algorithmic implementation is the **ID algorithm** [7].

Exercise

Use do-calculus to simplify the interventional query $P(Y|do(X))$ in the following graphs.





Recap: Causal Inference

Causal inference provides theory and methods to exploit graphs and data to reduce *interventional queries* to *observational queries*.

Intervention $\rightsquigarrow$ **Conditioning**

It creates a bridge:

$$\begin{tikzcd}\text{causality formalism} \arrow[r] & \text{statistical formalism} \\ \text{interventional domain} \arrow[r] & \text{observational domain}\end{tikzcd}$$

Thanks!

Thank you for listening!

References I

[1] Elias Bareinboim, Juan D Correa, Duligur Ibeling, and Thomas Icard. *On Pearl's Hierarchy and the Foundations of Causal Inference*. 2022.

[2] Mathias Frisch. *Causal reasoning in physics*. Cambridge University Press, 2014.

[3] Thomas F Icard. From programs to causal models. In *Proceedings of the 21st Amsterdam colloquium*, pages 35–44, 2017.

[4] Judea Pearl. *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2 edition, 2009.

[5] Judea Pearl. Causal diagrams for empirical research (with discussions). In *Probabilistic and causal inference: The works of Judea Pearl*, pages 255–316. 2022.

[6] Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. *Elements of causal inference: Foundations and learning algorithms*. MIT Press, 2017.

[7] Ilya Shpitser and Judea Pearl. Complete identification methods for the causal hierarchy. *Journal of Machine Learning Research*, 9(Sep):1941–1979, 2008.

[8] Ilya Shpitser and Judea Pearl. Identification of conditional interventional distributions. *arXiv preprint arXiv:1206.6876*, 2012.